

BGE-M3 Dense, Sparse, and Multi-Vector Representation

BGE-M3 Dense, Sparse, and Multi-Vector Representation

Document ID: doc_12_embeddings_bge_m3

Topics: embeddings, bge-m3, dense

1. Overview

This technical specification details bge-m3 dense, sparse, and multi-vector representation within the MCP PRO orchestration platform.

Detailed sections cover architectural contracts, runtime memory budgets, error handling semantics, and telemetry capture.

2. Implementation Details

The subsystem is encapsulated in the MCP PRO codebase, offering high-throughput, low-latency execution interfaces.

Default parameters are configured for enterprise stability and scalable vector retrieval.

3. Validation and Benchmarking Standards

All performance assertions must be validated through automated benchmark suites recording latency percentiles (mean, p50, p90, p95, p99), error rates, and resource utilization.